

ASP.NET + Angular Hosting & VM Recommendation

Professional Hosting Infrastructure Recommendation Report

This report provides a complete production hosting recommendation for the ASP.NET Core and Angular-based enterprise application after reviewing the provided SQL database structure and operational modules.

Project Overview

The project is an enterprise operational platform developed using ASP.NET Core backend, Angular frontend, and MySQL database architecture.

Database analysis indicates:

- Around 196 operational tables
- ERP/CRM style architecture
- HR and attendance modules
- Customer and order management
- Reporting and operational tracking
- Heavy API and database interaction

Recommended Hosting Architecture

Recommended deployment architecture:

User → Nginx → ASP.NET Core API → MySQL Database

Angular frontend will be hosted as a production static build using Nginx.

This architecture provides:

- Better performance

- Higher stability
- Easier scalability
- Improved security

Recommended Operating System

Ubuntu Server 24.04 LTS is strongly recommended.

Reasons:

- Lower server cost
- Better ASP.NET Core performance on Linux
- Lower RAM usage
- Better production stability
- Easier SSL and deployment configuration

Recommended VM Configuration

Resource	Recommended Configuration
CPU	4 vCPU
RAM	8 GB
Storage	120–160 GB NVMe SSD
Operating System	Ubuntu 24.04 LTS
Database	MySQL 8
Web Server	Nginx
Runtime	.NET 8
SSL	Let's Encrypt

Why This Configuration?

The application contains multiple operational modules, reporting systems, attendance tracking, customer workflows, and continuous data growth.

The recommended configuration ensures:

- Stable API performance

- Faster MySQL query execution
- Smooth Angular frontend performance
- Better concurrent user handling
- Future scalability without immediate infrastructure migration

Recommended Hosting Providers

Provider	Why Recommended	Approx Monthly Cost
Hetzner Cloud	Best price-to-performance ratio	■3000–■5000
DigitalOcean	Easy deployment and management	■4500–■7000
AWS EC2	Enterprise-grade scalability	■8000+
Contabo	Budget-friendly option	■2500–■4500

Production Recommendations

- Configure automatic daily database backup
- Enable HTTPS using SSL certificate
- Use firewall restrictions for production security
- Configure monitoring for CPU, RAM, and API response
- Separate development, staging, and production environments
- Maintain backup retention for at least 7–30 days

Final Recommendation

Recommended Provider: **Hetzner Cloud**

Recommended VM:

- 4 vCPU
- 8GB RAM
- 120–160GB NVMe SSD
- Ubuntu 24.04 LTS

This configuration provides the best balance between cost, performance, stability, and future scalability for the current project structure.

